

Experiment No. 2 — OLAP Operations

Aim: To study and perform OLAP operations such as Slice, Dice, Drill-down, Roll-up and Pivot on a multidimensional data set.

PART A — THEORY

1. What is OLAP?

OLAP stands for **Online Analytical Processing**.

OLAP is a technology used to analyze large amounts of data from different perspectives or dimensions.

It is mainly used in:

- Business Intelligence
- Data Warehousing
- Reporting
- Decision making
- Sales analysis
- Financial analysis
- Customer analysis
- Trend analysis

The main purpose of OLAP is not to record transactions, but to **analyze existing data**.

Simple example

Suppose a company has millions of sales transactions. Management may want to ask:

“How much did we sell in Mumbai?”

Then:

“How much did we sell in Mumbai for Electronics?”

Then:

“How much did we sell in Mumbai for Electronics during Q1?”

OLAP operations allow us to look at the same data from these different perspectives.

2. OLTP vs OLAP

OLTP — Online Transaction Processing

OLTP systems are used for **day-to-day transactions**.

- Placing an order
- Making a payment
- Booking a ticket
- Updating customer information
- Recording a sale

Customer buys Laptop
↓
Transaction recorded
↓
Payment processed
↓

Inventory updated

OLTP focuses on **individual transactions**.

OLAP — Online Analytical Processing

OLAP takes historical data and analyzes it.

Millions of Sales Transactions
↓
Data Warehouse
↓
OLAP
↓
Business Analysis
↓
"Which city has the highest sales?"

OLAP focuses on **analysis and aggregation**.

OLTP	OLAP
Transaction processing	Analytical processing
Current data	Historical data
Many small transactions	Large analytical queries
Insert/Update/Delete	Mostly Read
Highly normalized	Usually dimensional
Used by operational users	Used by analysts/managers
Example: Banking transaction	Example: Yearly sales analysis

3. OLAP Cube

The central concept behind OLAP is the **multidimensional data cube**.

A data cube represents data using multiple **dimensions** and **measures**.

Dimensions

Time
Product
Location

Measure

Sales Amount

Conceptually:

TIME
↑
|
|
|
+-----→ PRODUCT
/
/
/
LOCATION

Each combination of dimensions contains a measure.

For example:

Mumbai + Electronics + 2025
↓
■75,000

Pune + Furniture + 2025
↓
■45,000

The cube allows us to move through different levels of detail.

4. Dimensions

A **dimension** is a perspective or category through which data can be analyzed.

Time Dimension

Year
↓
Quarter
↓
Month
↓
Day

Location Dimension

Country
↓
State
↓
City
↓
Store

Product Dimension

Category
↓
Subcategory
↓
Product

For example:

Electronics
↓
Computers
↓
Laptop

Dimensions provide the **context** for measures.

5. Measures

A **measure** is a numerical value that can be analyzed or aggregated.

- SalesAmount
- Quantity
- Profit
- Revenue
- Cost
- Discount

Product	Quantity	Sales
Laptop	5	■2,75,000
Mouse	20	■16,000
Keyboard	10	■15,000

Here:

Quantity → Measure
Sales → Measure

6. Dimension Hierarchy

A dimension hierarchy represents different levels of detail.

For example, Time:

Year



Quarter



Month



Day



More accurately:

2025



Q1



January



February



March



Q2



April



May



June



This hierarchy is extremely important for **Drill-down and Roll-up**.

Drill-down

Year → Quarter → Month → Day

More detail.

Roll-up

Day → Month → Quarter → Year

More summarized.

7. OLAP Operations

The five important OLAP operations in this experiment are:

- Slice
- Dice
- Drill-down
- Roll-up
- Pivot

They all analyze the same underlying data but from different perspectives and levels of detail.

8. Slice Operation

Definition

Slice is an OLAP operation that selects a **single value from one dimension**, producing a smaller view of the data.

In simple words:

Fix one dimension to one particular value and examine the remaining data.

Example

Suppose we have:

Year
City
Product
Sales

If we select:

Year = 2025

we have sliced the cube at 2025.

3D Cube

```
Year
|
| 2024
| 2025 ← SELECT
| 2026
|
+-----
```

The result is a smaller dataset containing only 2025.

Important point: Slice generally selects one fixed value from one dimension.

Example:

Year = 2025

not:

Year = 2025 AND City = Mumbai

That starts becoming a **Dice** operation because multiple dimensions are being restricted.

9. Dice Operation

Definition

Dice selects a **sub-cube** by applying conditions to multiple dimensions.

In simple words:

Dice selects specific values or ranges from two or more dimensions.

Example

Year = 2025
City = Mumbai OR Pune

Product Category = Electronics

We have restricted multiple dimensions:

Year → 2025

City → Mumbai, Pune

Category → Electronics

This produces a smaller sub-cube.

Slice vs Dice

Slice	Dice
Usually one dimension is fixed	Multiple dimensions are restricted
Produces a slice of cube	Produces a sub-cube
Example: Year = 2025	Example: Year = 2025, City = Mumbai/Pune, Category = Electronics

Easy way to remember:

Slice = one cut

Dice = multiple cuts

10. Drill-down Operation

Definition

Drill-down moves from higher-level summarized data to lower-level detailed data.

It means:

Going from general information to more detailed information.

Time example

Year
↓
Quarter
↓
Month
↓
Day

Suppose initially we have:

2025 → ■10,00,000

Drill down:

2025
↓
Q1 → ■2,50,000
Q2 → ■3,00,000
Q3 → ■2,00,000
Q4 → ■2,50,000

Drill down again:

Q1
↓
January → ■80,000
February → ■90,000
March → ■80,000

So:

Drill-down = more detail.

11. Roll-up Operation

Definition

Roll-up is the opposite of drill-down.

It moves from detailed data to summarized data.

Day
↓
Month
↓
Quarter
↓
Year

Suppose we have:

Month	Sales
January	■80,000
February	■90,000
March	■80,000

Roll-up:

Q1 → ■2,50,000

Roll-up can also be performed across geographical hierarchies.

Store
↓
City
↓
State
↓
Country

So:

Store-level sales
↓
City-level sales
↓
State-level sales
↓
Country-level sales

Important difference

DRILL-DOWN
Less detail → More detail

ROLL-UP
More detail → Less detail

12. Pivot Operation

Definition

Pivot changes the orientation or arrangement of dimensions in an OLAP report.

It does **not necessarily change the underlying data**.

It simply changes how the information is displayed.

Before Pivot

City	Laptop	Mouse
Mumbai	55000	1600
Pune	70000	2400
Nashik	45000	800

After pivot

Product	Mumbai	Pune	Nashik
Laptop	55000	70000	45000
Mouse	1600	2400	800

The information is the same. The rows and columns have been exchanged.

Simple definition: Pivot = rotate/rearrange the dimensions of the report.

13. Summary of OLAP Operations

Operation	What it does	Example
Slice	Selects one value of a dimension	Year = 2025
Dice	Selects values from multiple dimensions	2025 + Mumbai/Pune + Electronics
Drill-down	Moves to more detail	Year → Quarter → Month
Roll-up	Moves to higher summary	Day → Month → Quarter → Year
Pivot	Changes report orientation	Rows ↔ Columns

PART B — COMMON EXAMPLE DATA

Now let's use **one table for every operation**.

14. Sales Table

Year	Quarter	Month	City	Product	Category	Sales
2025	Q1	Jan	Mumbai	Laptop	Electronics	55000
2025	Q1	Jan	Pune	Laptop	Electronics	60000
2025	Q1	Feb	Mumbai	Mouse	Electronics	1600
2025	Q1	Feb	Pune	Mouse	Electronics	2400
2025	Q1	Mar	Mumbai	Chair	Furniture	4500
2025	Q1	Mar	Pune	Chair	Furniture	5000
2025	Q2	Apr	Mumbai	Laptop	Electronics	55000
2025	Q2	Apr	Pune	Laptop	Electronics	60000
2025	Q2	May	Mumbai	Mouse	Electronics	1600
2025	Q2	May	Pune	Mouse	Electronics	2400
2025	Q2	Jun	Mumbai	Chair	Furniture	4500
2025	Q2	Jun	Pune	Chair	Furniture	5000
2026	Q1	Jan	Mumbai	Laptop	Electronics	65000
2026	Q1	Jan	Pune	Laptop	Electronics	70000
2026	Q1	Feb	Mumbai	Mouse	Electronics	1800
2026	Q1	Feb	Pune	Mouse	Electronics	2500

For simplicity, we will analyze the **Sales** column as our measure.

PART C — PERFORMING OLAP OPERATIONS

15. Slice Operation

Let's perform:

Slice where Year = 2025

We fix the **Year dimension** to 2025.

Original

2025
2026

We select:

Year = 2025

Result

Year	Quarter	Month	City	Product	Category	Sales
2025	Q1	Jan	Mumbai	Laptop	Electronics	55000
2025	Q1	Jan	Pune	Laptop	Electronics	60000
2025	Q1	Feb	Mumbai	Mouse	Electronics	1600
2025	Q1	Feb	Pune	Mouse	Electronics	2400
2025	Q1	Mar	Mumbai	Chair	Furniture	4500
2025	Q1	Mar	Pune	Chair	Furniture	5000
2025	Q2	Apr	Mumbai	Laptop	Electronics	55000
2025	Q2	Apr	Pune	Laptop	Electronics	60000
2025	Q2	May	Mumbai	Mouse	Electronics	1600
2025	Q2	May	Pune	Mouse	Electronics	2400
2025	Q2	Jun	Mumbai	Chair	Furniture	4500
2025	Q2	Jun	Pune	Chair	Furniture	5000

What happened?

We removed all 2026 records.

Original Data

↓

Year = 2025

↓

2025 Data Only

Therefore:

Slice = fixing one dimension to one value.

16. Dice Operation

Now let's perform a **Dice**.

Year = 2025

City = Mumbai or Pune

Category = Electronics

We're restricting **three dimensions**: Year, City and Category.

Result

Year	Quarter	Month	City	Product	Category	Sales
2025	Q1	Jan	Mumbai	Laptop	Electronics	55000
2025	Q1	Jan	Pune	Laptop	Electronics	60000
2025	Q1	Feb	Mumbai	Mouse	Electronics	1600
2025	Q1	Feb	Pune	Mouse	Electronics	2400
2025	Q2	Apr	Mumbai	Laptop	Electronics	55000
2025	Q2	Apr	Pune	Laptop	Electronics	60000
2025	Q2	May	Mumbai	Mouse	Electronics	1600
2025	Q2	May	Pune	Mouse	Electronics	2400

Notice that:

- 2026 → removed
- Furniture → removed
- Mumbai/Pune → retained

Therefore:

Dice creates a smaller sub-cube by applying conditions on multiple dimensions.

17. Drill-down Operation

Now let's say management initially wants **sales by year**.

We aggregate our data.

Level 1 — Year

For 2025:

$55000 + 60000 + 1600 + 2400 + 4500 + 5000$
 $+ 55000 + 60000 + 1600 + 2400 + 4500 + 5000$
 $= \blacksquare 260,000$

For 2026:

$65000 + 70000 + 1800 + 2500$
 $= \blacksquare 139,300$

Year	Total Sales
2025	$\blacksquare 260,000$
2026	$\blacksquare 139,300$

This is our high-level view.

Drill down: Year → Quarter

Year	Quarter	Total Sales
2025	Q1	■128,500
2025	Q2	■131,500
2026	Q1	■139,300

Now we can see which quarter generated the sales.

Drill down again: Quarter → Month

Year	Quarter	Month	Sales
2025	Q1	Jan	■115,000
2025	Q1	Feb	■4,000
2025	Q1	Mar	■9,500

So our hierarchy is:

Year
 ↓
 Quarter
 ↓
 Month
 ↓
 Day

Every downward movement gives us **more detailed information**.

Therefore:

Drill-down = moving from summarized data toward detailed data.

18. Roll-up Operation

Now do exactly the opposite.

Suppose we start with monthly data:

Year	Quarter	Month	Sales
2025	Q1	Jan	■115,000
2025	Q1	Feb	■4,000
2025	Q1	Mar	■9,500
2025	Q2	Apr	■115,000
2025	Q2	May	■4,000
2025	Q2	Jun	■9,500

We roll up:

Month
↓
Quarter

Result

Year	Quarter	Sales
2025	Q1	■128,500
2025	Q2	■131,500

Then roll up again:

Quarter
↓
Year

Result

Year	Total Sales
2025	■260,000

So:

DAY
↓
MONTH
↓
QUARTER
↓
YEAR

is a **roll-up hierarchy**.

Therefore:

Roll-up = moving from detailed data toward summarized data.

19. Pivot Operation

Now let's analyze sales by city and product.

Initially

City	Product	Sales
Mumbai	Laptop	55000
Pune	Laptop	60000
Mumbai	Mouse	1600
Pune	Mouse	2400
Mumbai	Chair	4500

City	Product	Sales
Pune	Chair	5000

Now we **pivot** the data.

Before Pivot

City → Product → Sales

After Pivot

City	Laptop	Mouse	Chair
Mumbai	■55,000	■1,600	■4,500
Pune	■60,000	■2,400	■5,000

So the resulting report becomes:

City	Laptop	Mouse	Chair
Mumbai	■55,000	■1,600	■4,500
Pune	■60,000	■2,400	■5,000

Now let's reverse the orientation.

Pivot again

Product	Mumbai	Pune
Laptop	■55,000	■60,000
Mouse	■1,600	■2,400
Chair	■4,500	■5,000

The data itself hasn't changed. Only its **orientation/presentation** changed.

20. All Five Operations Together

```

OLAP
|
+-----+-----+
| | |
SLICE DICE PIVOT
| | |
One value Multiple Change
of one dimensions orientation
dimension restricted
|
|
+-----+
|
DRILL-DOWN
↑
More detail
|
Year → Quarter
|
Quarter
↓
Month → Day

ROLL-UP
↑
More summary
|
Day → Month
|
Month → Quarter
|
Quarter → Year

```

21. Final Comparison

Operation	Direction / Action	Example
Slice	Fix one dimension	Year = 2025
Dice	Filter multiple dimensions	2025 + Mumbai/Pune + Electronics
Drill-down	More detail	Year → Quarter → Month
Roll-up	More summary	Month → Quarter → Year
Pivot	Change orientation	Rows ↔ Columns

One-line memory trick

Slice = one filter

Dice = multiple filters

Drill-down = zoom in

Roll-up = zoom out

Pivot = rotate the view

22. Result

The OLAP operations **Slice, Dice, Drill-down, Roll-up and Pivot** were studied and applied to a Retail Sales dataset. These operations allow the same data to be analyzed from different dimensions and levels of detail, making OLAP useful for **business intelligence, reporting and decision-making**.